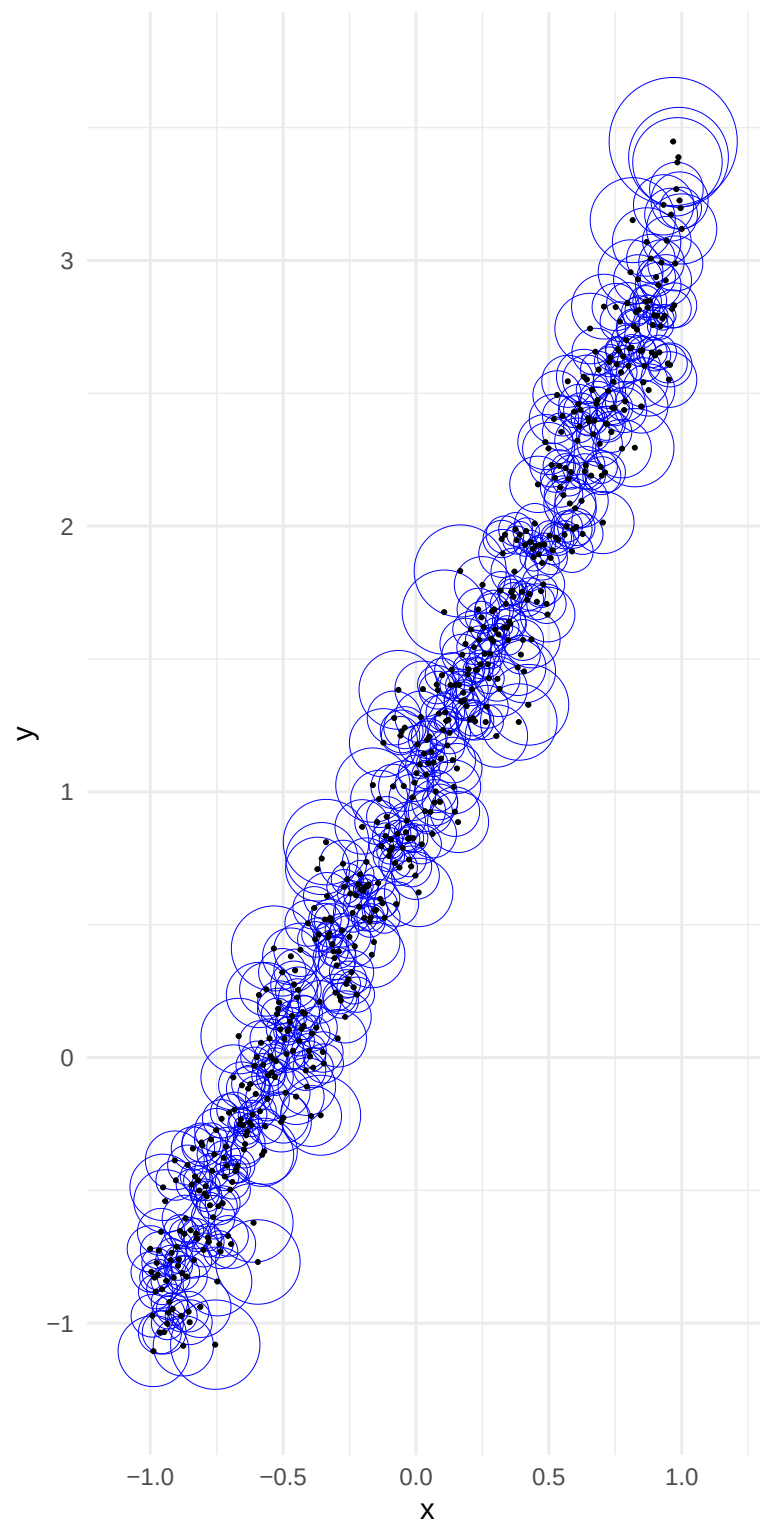
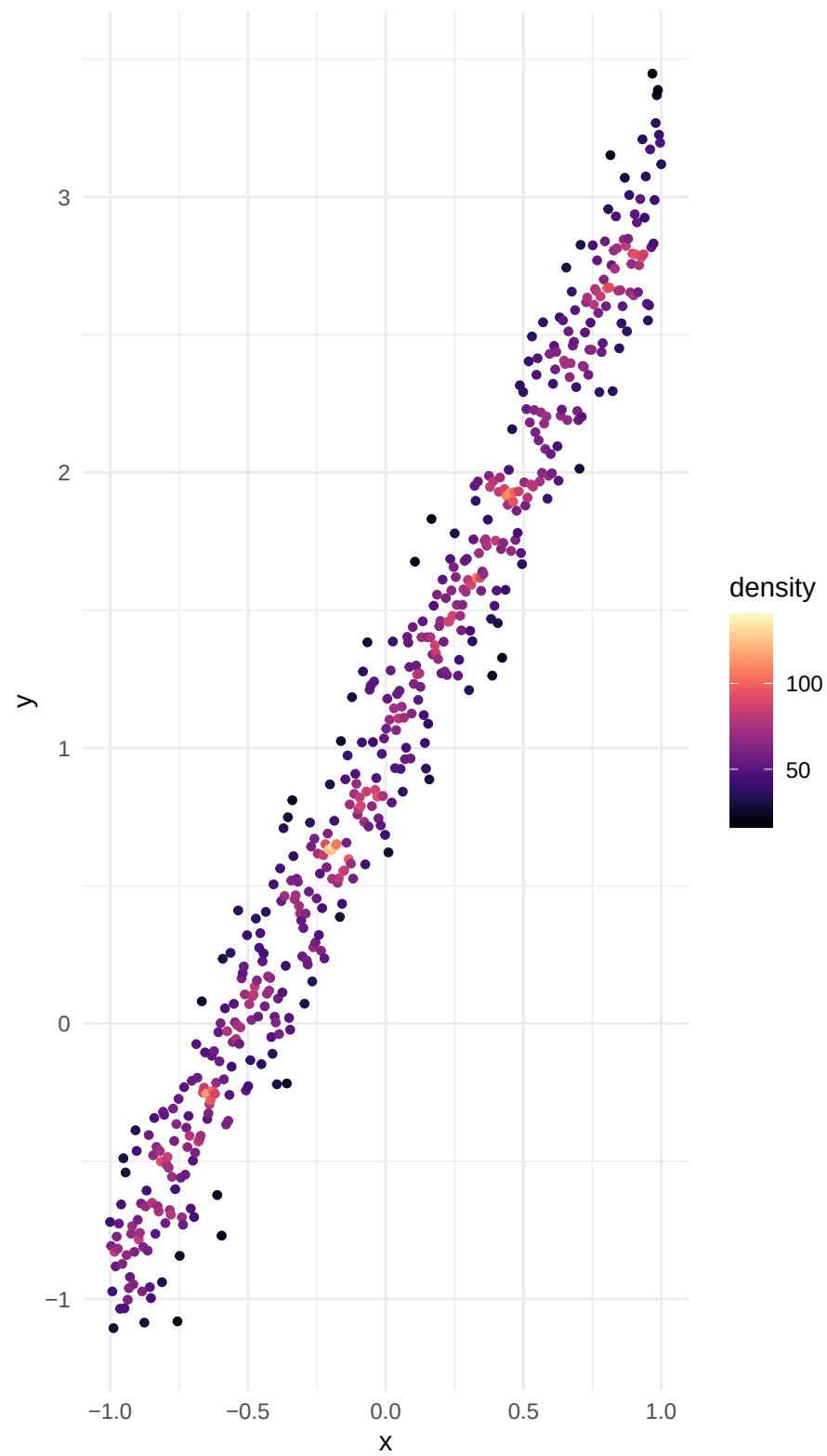


All Adaptive Spheres

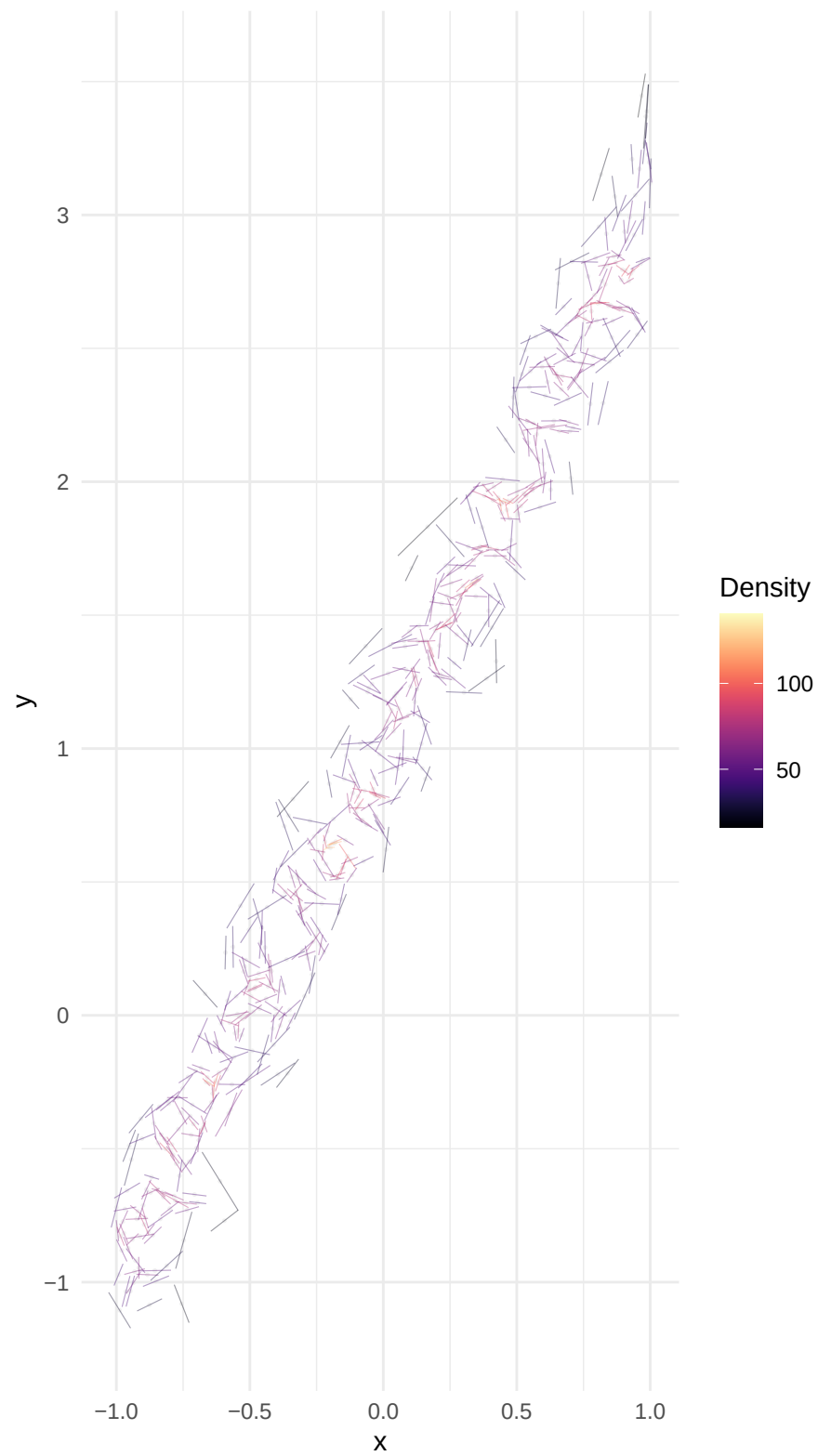
Visualizing local neighborhood size (radius) for every point.



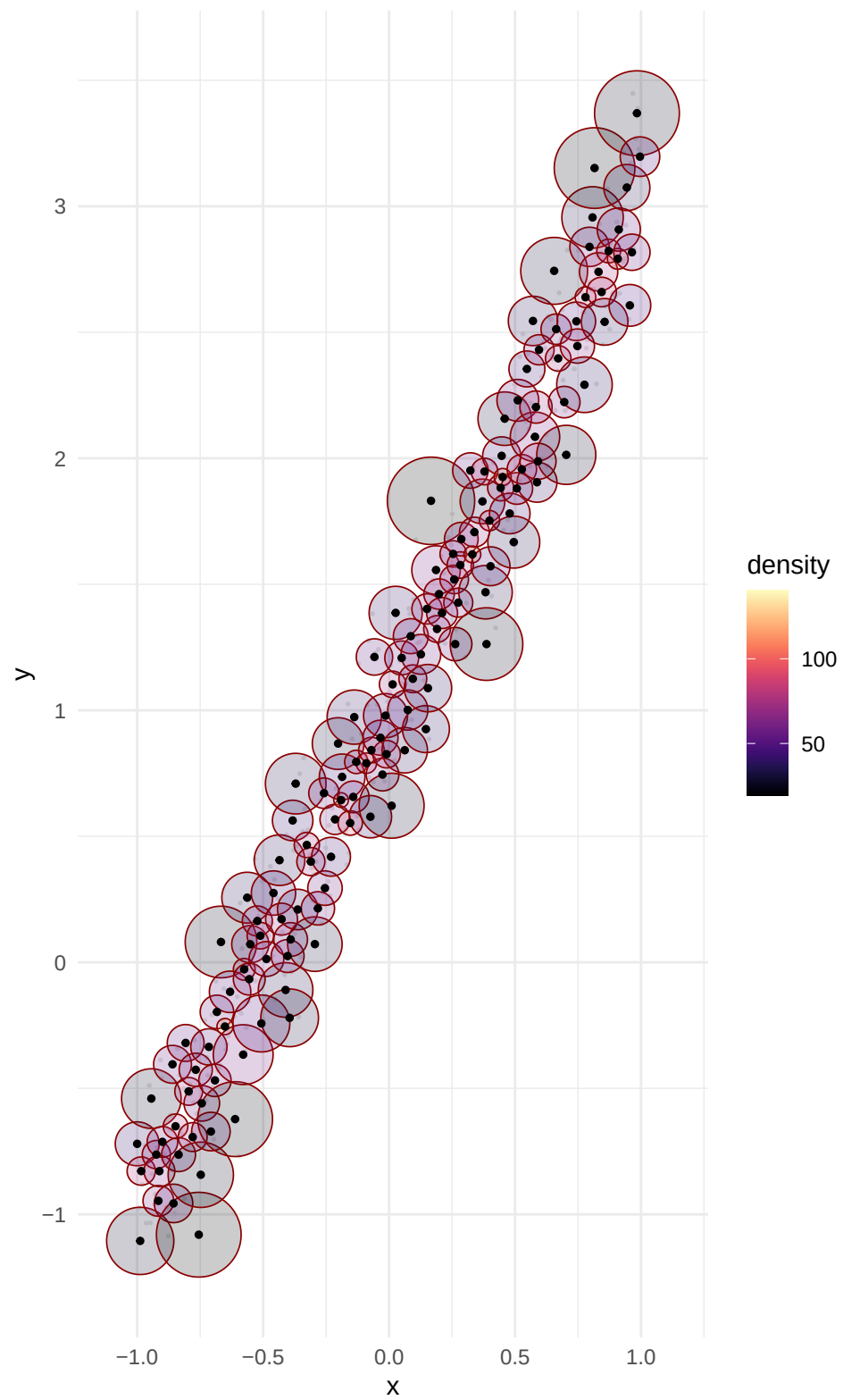
Point Density Distribution



Global Tangent Field (PC1)

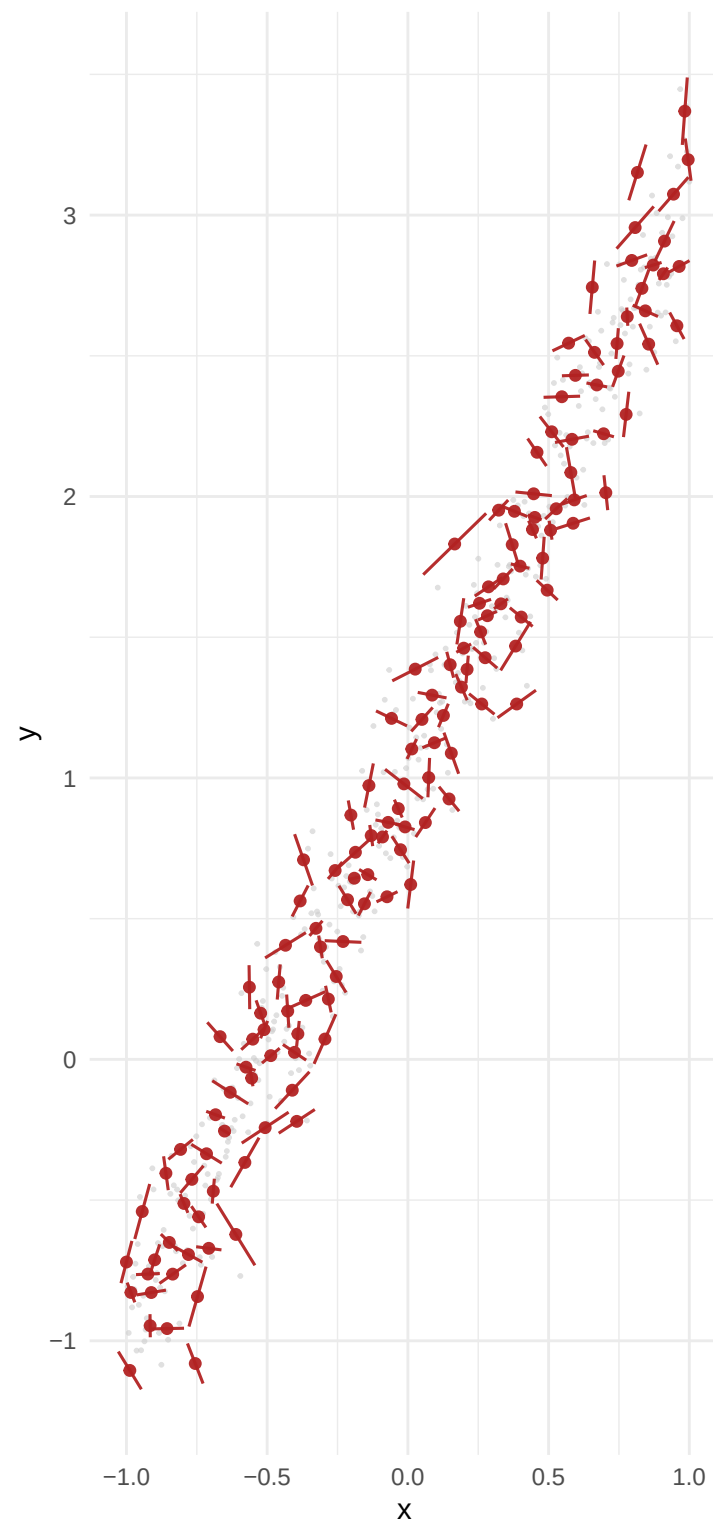


Atlas Selection (Anchors)



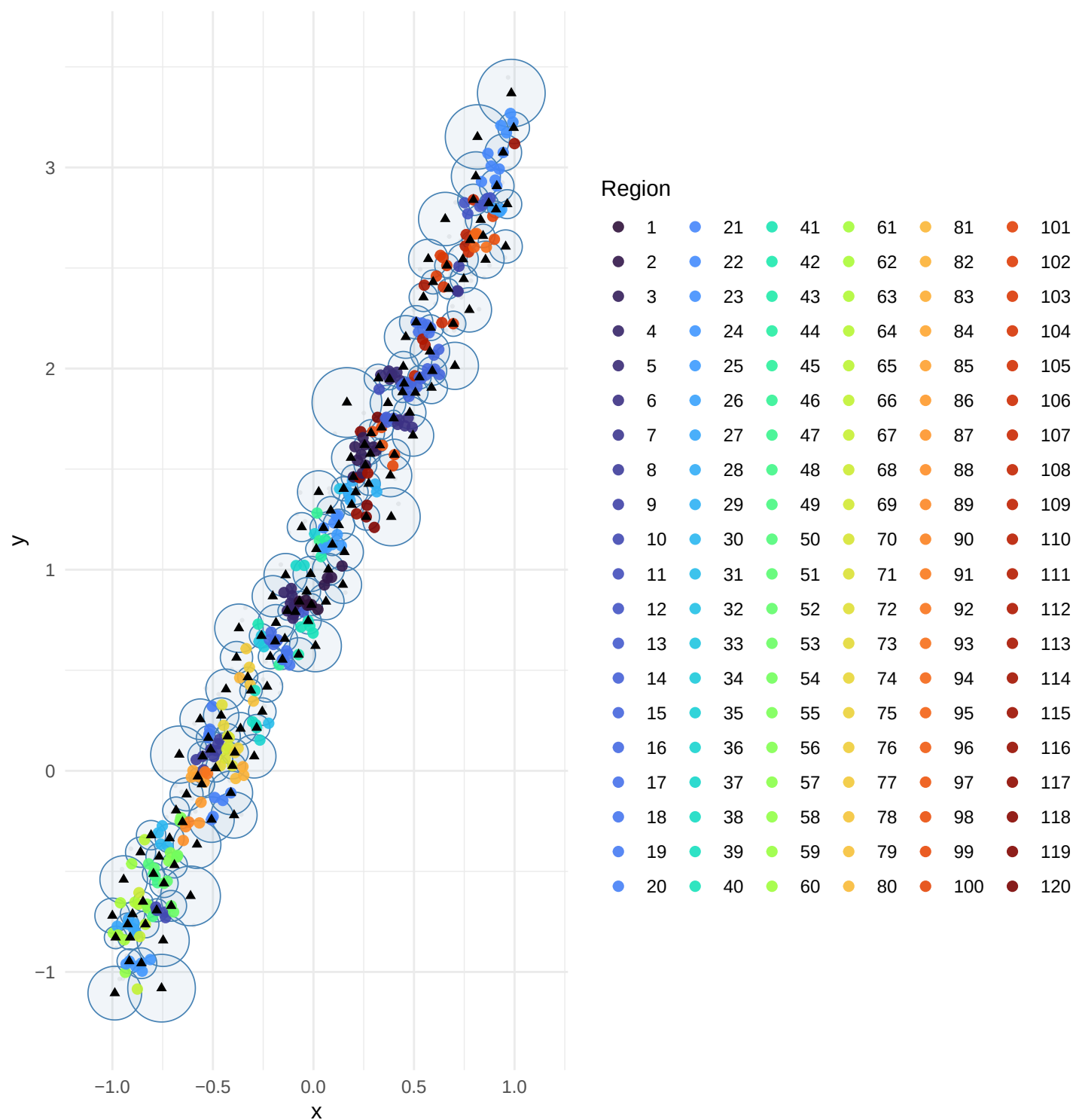
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



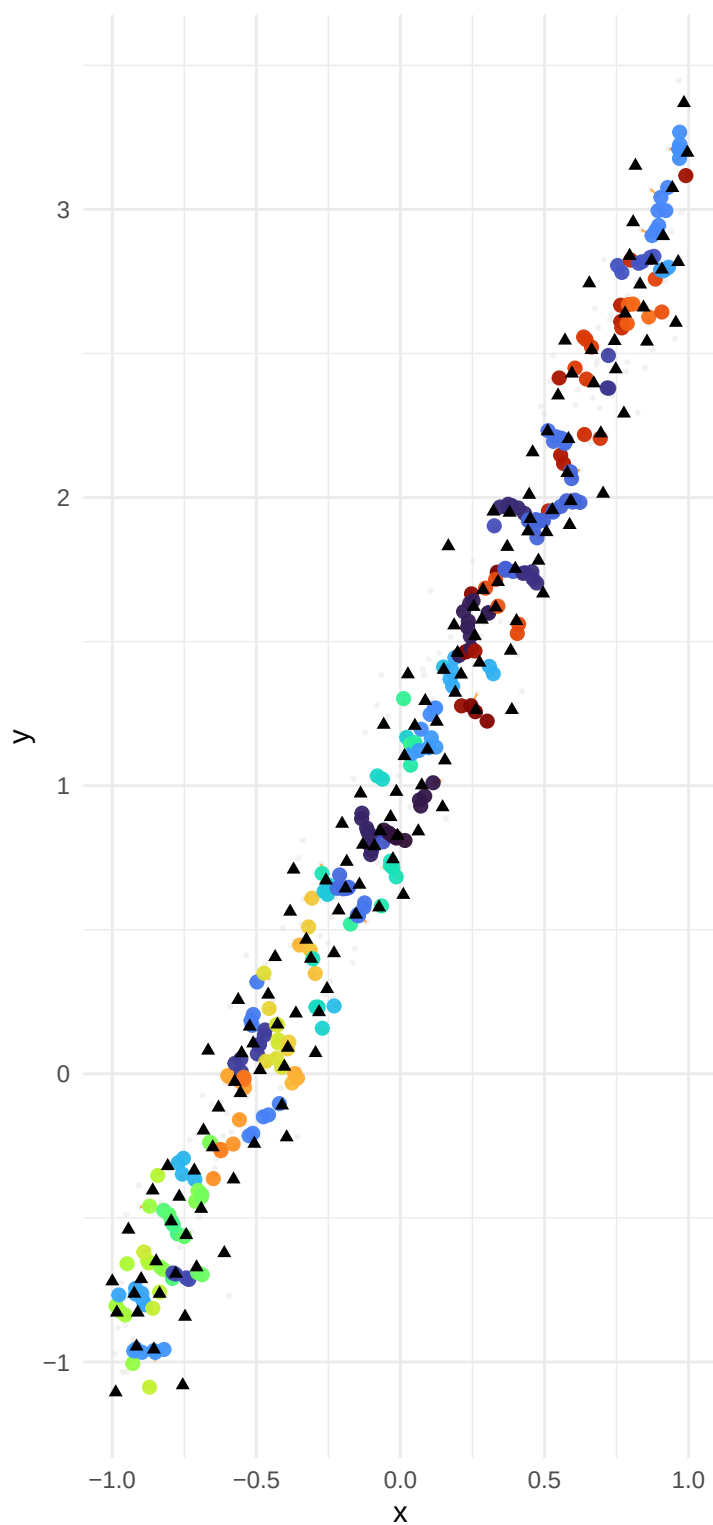
Intersection Regions (2+ anchors)

Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors

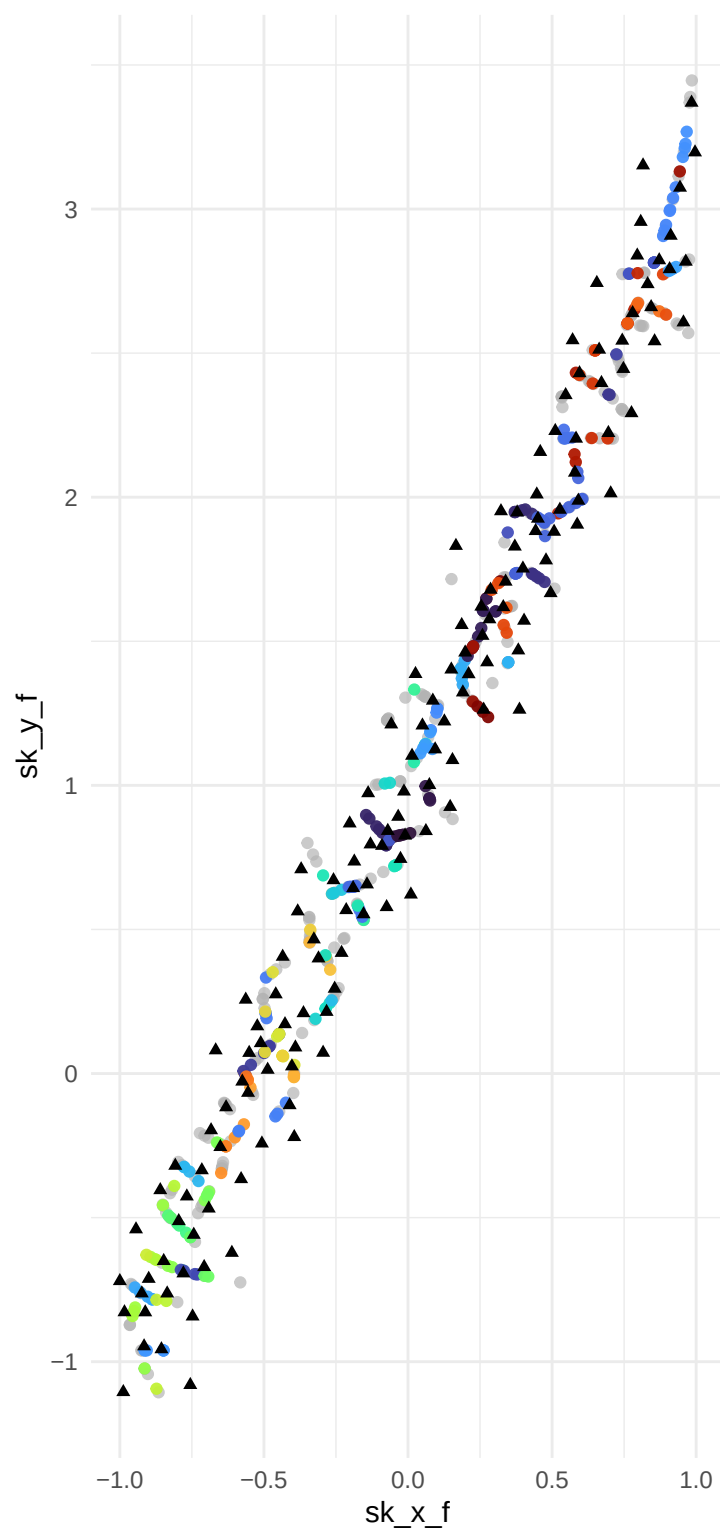


Region

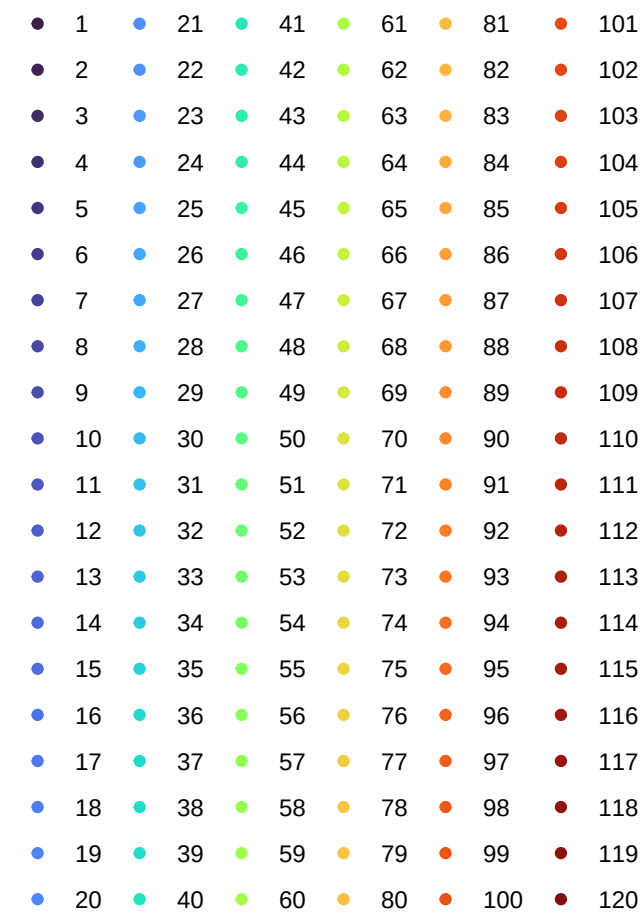
1	21	41	61	81	101
2	22	42	62	82	102
3	23	43	63	83	103
4	24	44	64	84	104
5	25	45	65	85	105
6	26	46	66	86	106
7	27	47	67	87	107
8	28	48	68	88	108
9	29	49	69	89	109
10	30	50	70	90	110
11	31	51	71	91	111
12	32	52	72	92	112
13	33	53	73	93	113
14	34	54	74	94	114
15	35	55	75	95	115
16	36	56	76	96	116
17	37	57	77	97	117
18	38	58	78	98	118
19	39	59	79	99	119
20	40	60	80	100	120

Final Stitched Backbone (Converged)

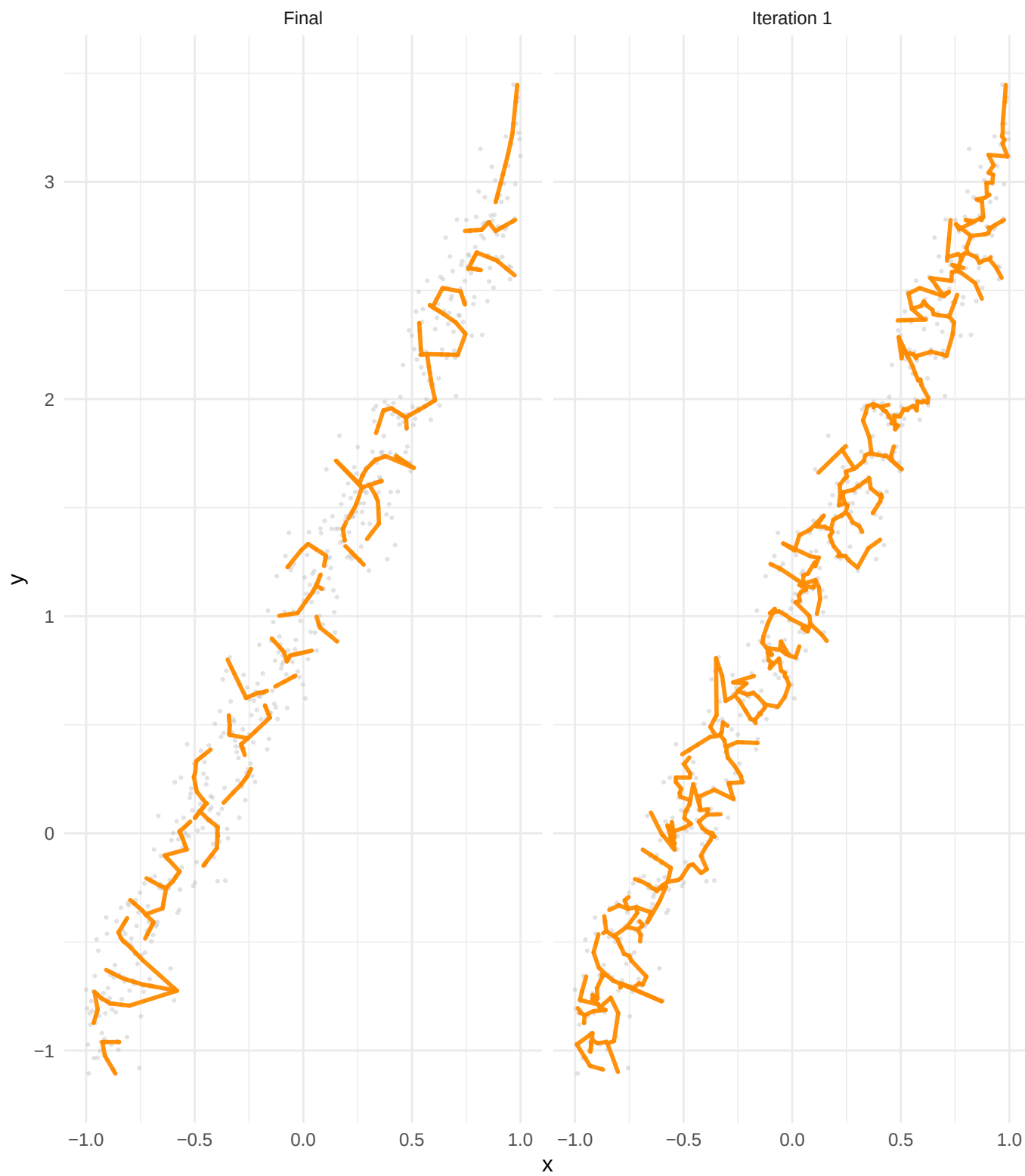
Final stitched points



Region

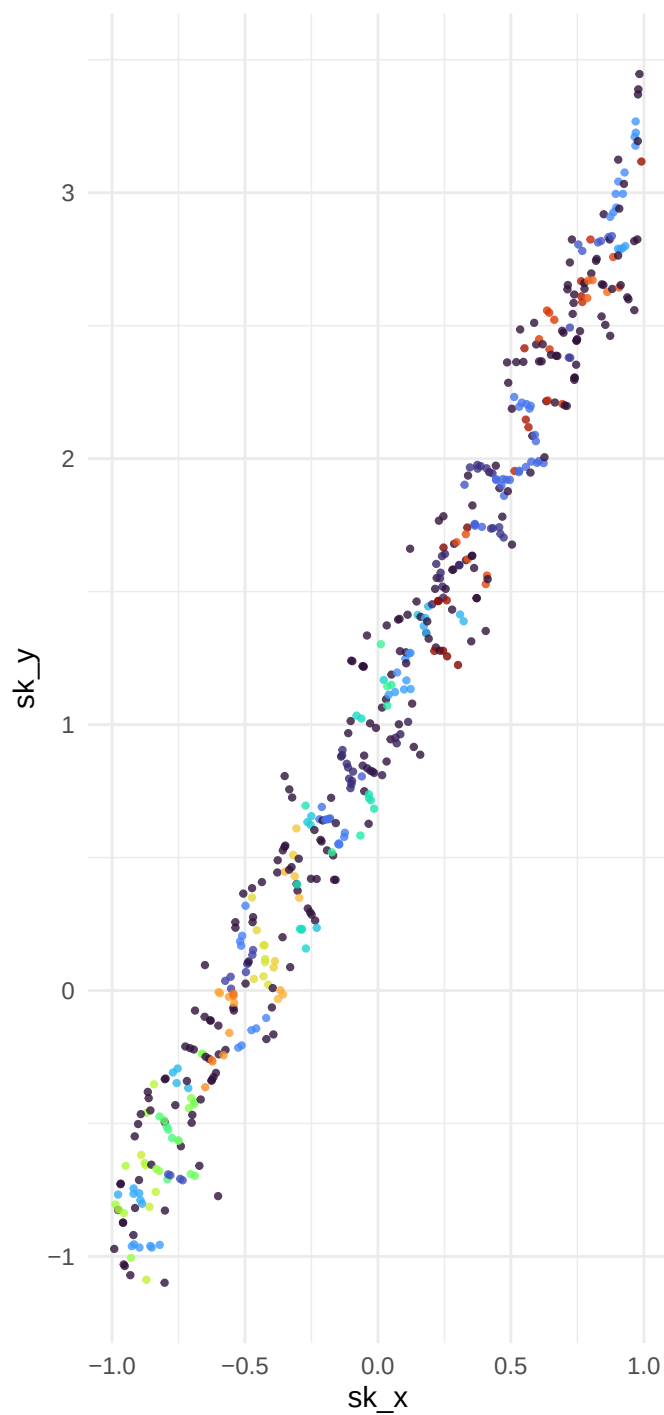


Backbone Edges

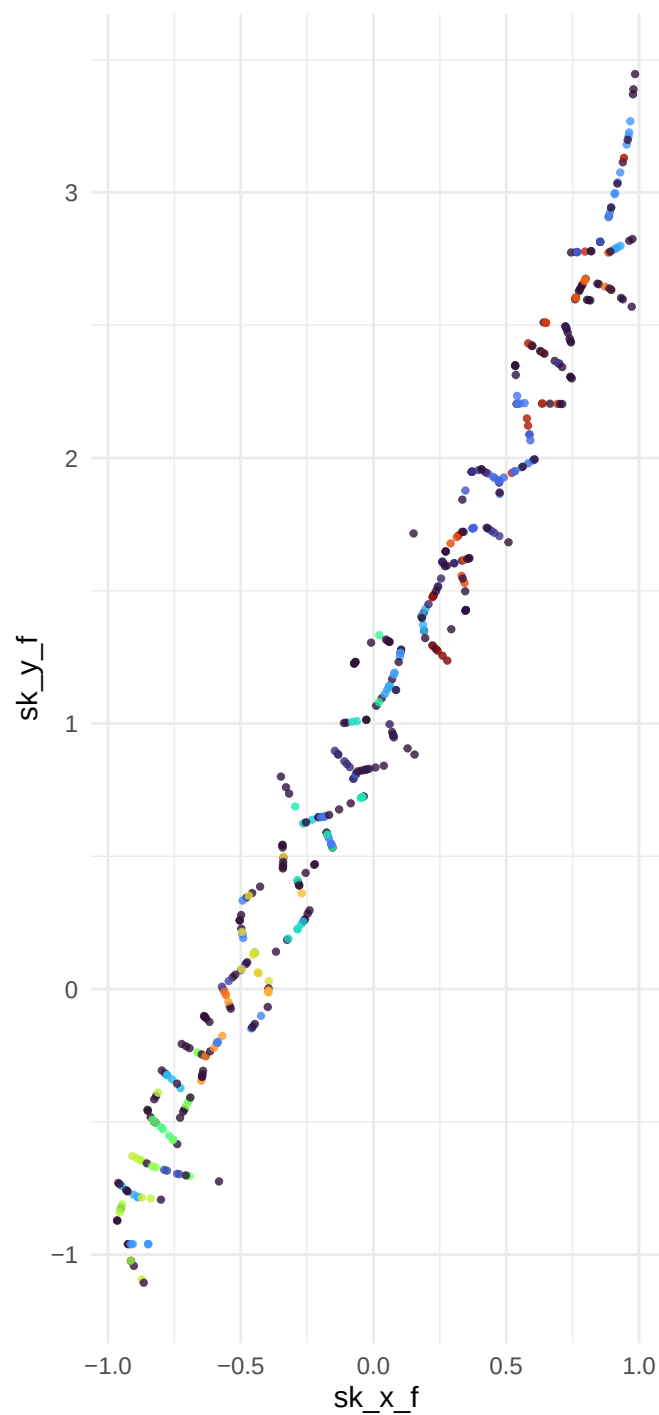


Convergence: iteration 1 vs final

Iteration 1

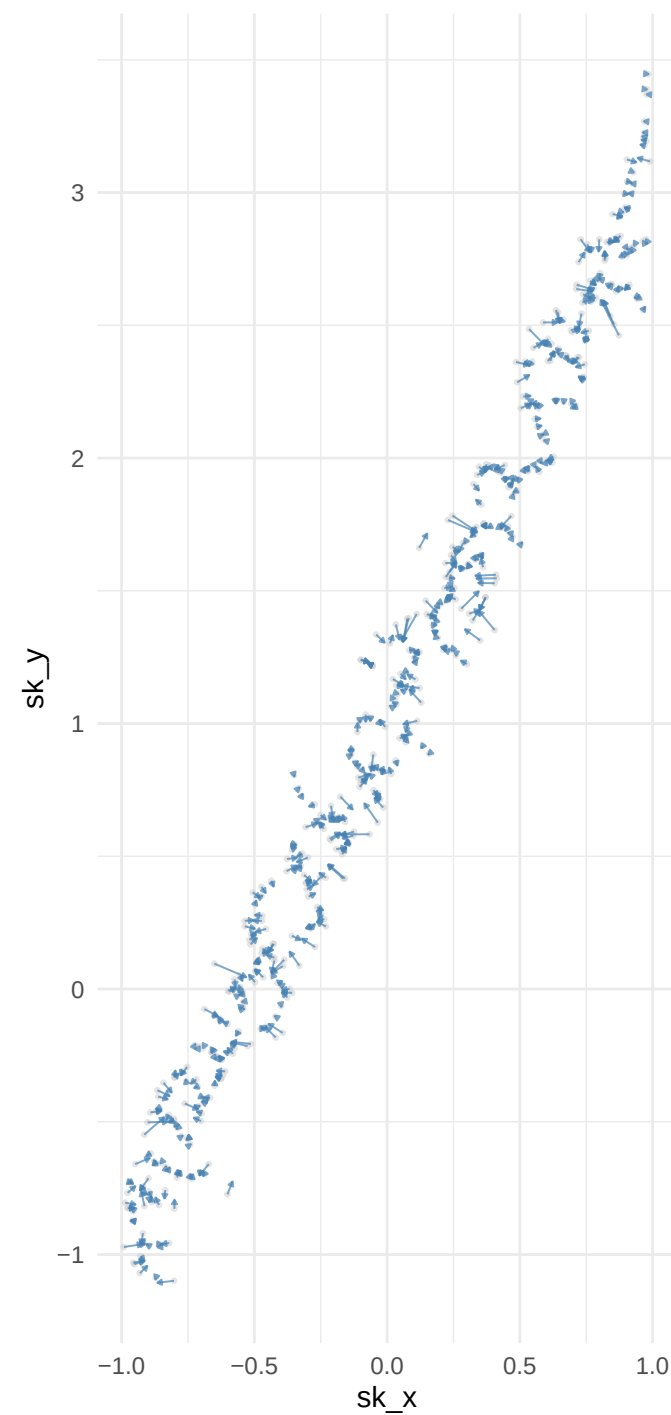


Converged



Displacement iter1 $\rightarrow$ final

Points moved $> 1e-6$: 500 / 500



Parameters: $k_{min} = 5$, $g_{threshold} = 3.0$, $o_{max} = 0.3$, $o_{min} = 0.10$, $stability = 0.90$, $scale_factor = 2.5$, $max_iter = 50$, $conv_tol = 0.01$